

2 Dars.

1 Darsda biz end to end LLM yaratishni o'rganishda ma'lumot yig'ishdan boshlab to'xtanish model arxitekturasini, pretraining, evaluation, inference va deploymentgacha bo'lgan jarayonni ko'rib chiqib oldik.

2 Darsda, shu katta pipeline'ning eng boshqariladigan va eng muhim qismini training data engineering ga diqqat qaratib o'tiramiz.

LLM uchun shunchaki juda ko'p ma'lumot topish yetarli emas. Bu ma'lumotning sifat manbasi, litsenziyasi, va aralashtirishini ham muhandislik asosida boshqarish kerak.

Shunday qilib, 1. katta dataset yetarli emas. Modeling xatoqidagi bilim til xatolari, to'g'ri va noto'g'ri va qabuliyatli training data o'zgarishi shartlarini.

Model arxitekturasini inson miya tezligiga bo'lsa, training data, uning barcha qismlari, hayot o'quvini ko'rgan o'qigan va eshitgan tajribasidir.

Agar inson noto'g'ri ma'lumat o'qish kuchi xotiraga bo'lishining o'zi uni bilimli qilmaydi. Xudoli shunday katta transformer ham sig'irsiz dataset bilan modelga aylanmaydi.

Loyihani xinalaylash $\Rightarrow$ Data topish $\Rightarrow$

$\Rightarrow$ Sifatini tekshirish $\Rightarrow$ Data mixture

Yaratish $\Rightarrow$ Etsenziya yaratish $\Rightarrow$

Muhandislik garori.

- gitignore $\rightarrow$ git tomonidan yaratilmasligi kerak bo'lgan fayl va papkalarni uchun yoziladigan qoidalar fayli.

Git hubga push qilinishi kerak bo'lgan fayllar ro'yxati: etarici aytiladi.

katta model weightlar

data setlar

maxfiy to'ra API.

Data Source.

training uchun ma'm gayerdan olinganligi.

Bilolizordi, lekin data source va tayyor

training dataset bir xil narxa emas.

Masalan

Common Crawl → xon web manbasi

Fine Web → Common Crawl asosida qaytariq

HF → dataset saqlash va taqsimot.

Dadma - taxvor corpus hamisha uni qaytadan
bilar bog'liq vositalar.

Common Crawl internet saxifalarining
kattik arxividir. Unda web crawl natijalar
bir nechta formatda bo'lishi mumkin.

WARC, WAT, WET

Common Crawl Internet oqimida tashkil
topilgan to'g'ri o'rta. Unda bo'liq xon bor
o'qatxan bor.

Fine web → Common Crawl dan olingan web
materiallarni extraction, filtring, deduplication,
bosqichlarni o'tirish o'tirish corpus.

Lexicon tashkil etish, lexic, lexic, lexic, lexic, lexic,
gumori lexic, lexic, lexic, lexic, lexic.

Dolma Allen Institute for AI
data processing etotizimulir.

General Web $\Rightarrow$ Common Crawl Fine Web
Reference data Wikipedia, encyclopedia
Scientific data arXiv, arXiv, arXiv
Education data Fine Web data
Code data $\Rightarrow$ Licensia, arXiv, Git repo
Code data test variables.

Data Engineering flat data.

Domain of definition $\Rightarrow$ Source discovery $\Rightarrow$
 $\Rightarrow$ License review $\Rightarrow$ Collection $\Rightarrow$ Persisting
 $\Rightarrow$ Norm

Quality Dimension bilim
Data quality bita son emas. U bilim
o'chiruvchi bilimlarni natijalar.

$Q(D) = f(\text{relevance, accuracy, diversity,}$
 $\text{security, integrity, legality, provenance})$

Katta dataset != Yaxshi model

Common Crawl $\Rightarrow$ Cleaned $\Rightarrow$ de duplicated
 $\Rightarrow$ Fine web $\Rightarrow$ $\Rightarrow$ $\Rightarrow$ Jstor

Bu ~~gala~~ muxim savol. Cleaning va de duplication. Tora data bolishi mumkin
Cekin model ~~maqsadiga~~ alogasi bo'lsa
ligi mumkin.

Agar cleaning model 35% yangilik
va ijtimoiy sharh matnida o'qitilgan
data tash bo'lsa ham model kerakli
o'stinchiga aylanmaydi.